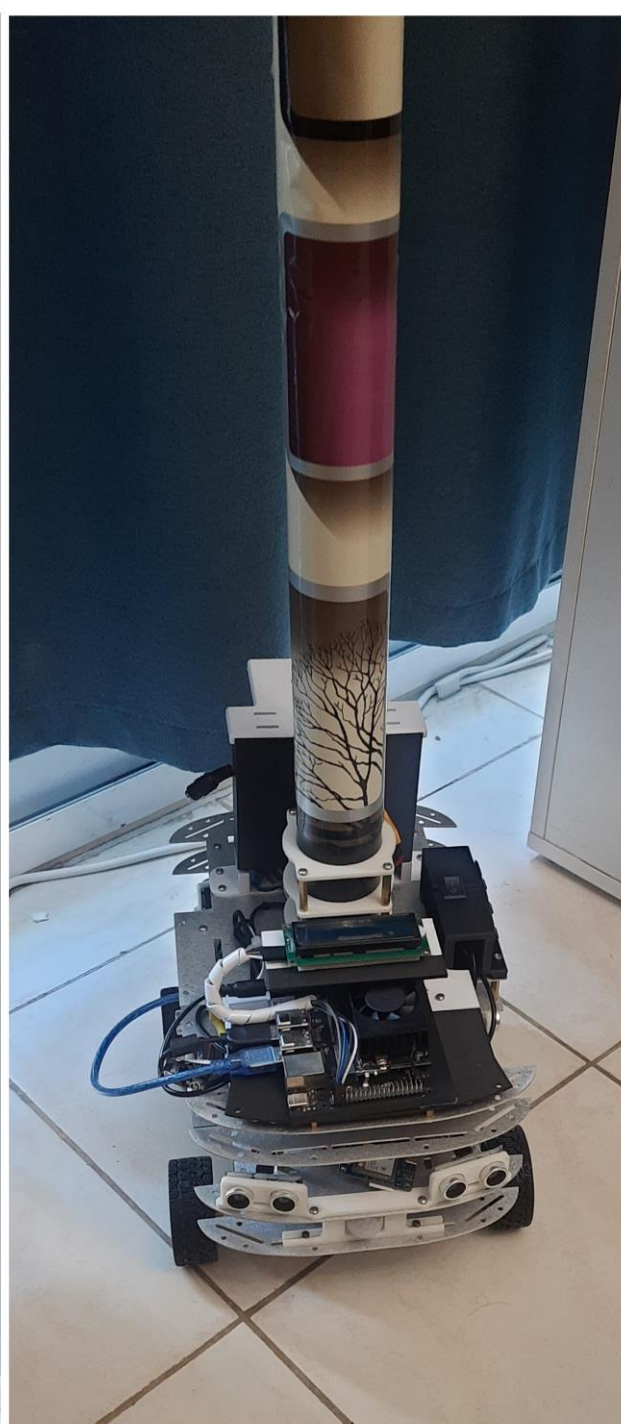


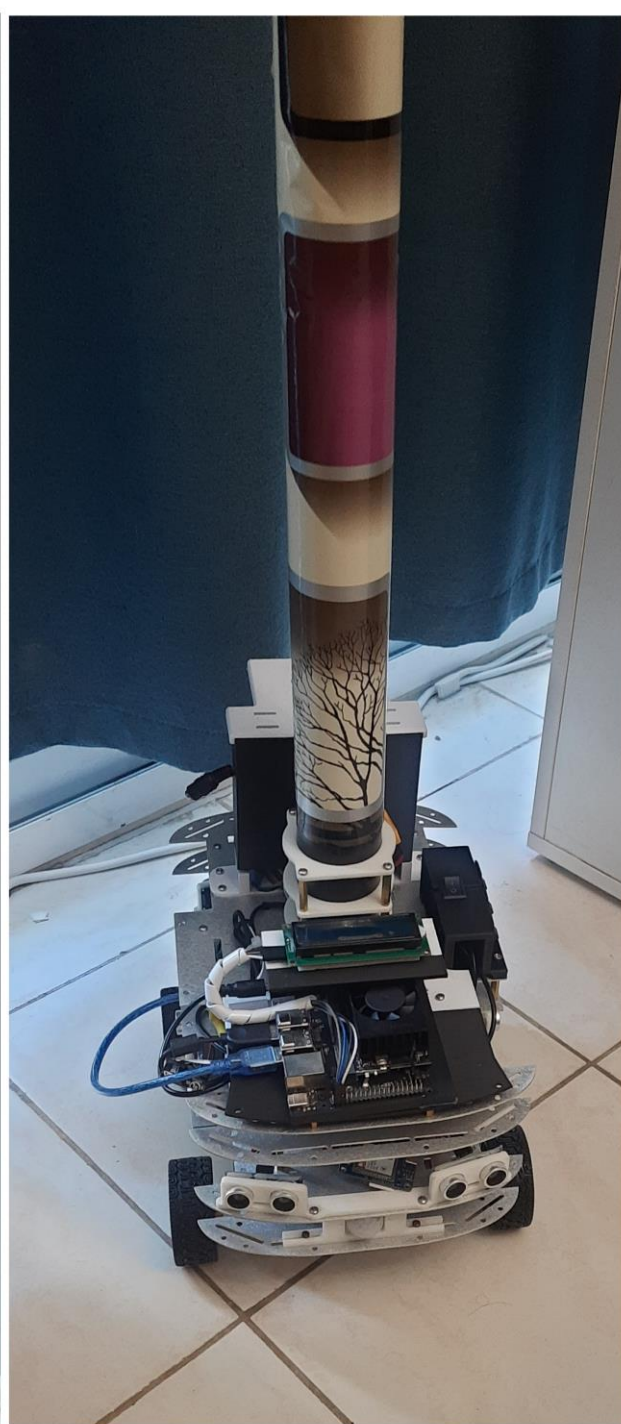


CARA – Companion Assistive Robot for the Aged

A low-cost GSM-based assistive robot for elderly care in developing countries

PDE4445 Robotics Dissertation Project

- Jayashanka Anushan (M01037028)
- Supervisor: Professor Judhi Prasetyo



MOTIVATION

The Care Gap in Developing Countries

A Growing Crisis

Rapidly ageing populations with insufficient infrastructure – many elderly live alone without consistent care.

Existing Solutions Fall Short

PARO, Pepper, and Care-O-Bot cost thousands of dollars and require Wi-Fi – neither affordable nor deployable in low-resource settings.

The Design Imperative

A sub-\$500, GSM-connected robot with onboard AI – viable where current systems are not.

Five Research Questions

1

Meaningful Assistive Support

Can a low-cost robot provide meaningful assistive support?

2

Onboard Vision for Fall Detection

Can low-cost onboard vision achieve reliable fall detection?

3

GSM Connectivity

Can GSM replace Wi-Fi-dependent connectivity for safety alerts?

4

Cost-Driven Trade-offs

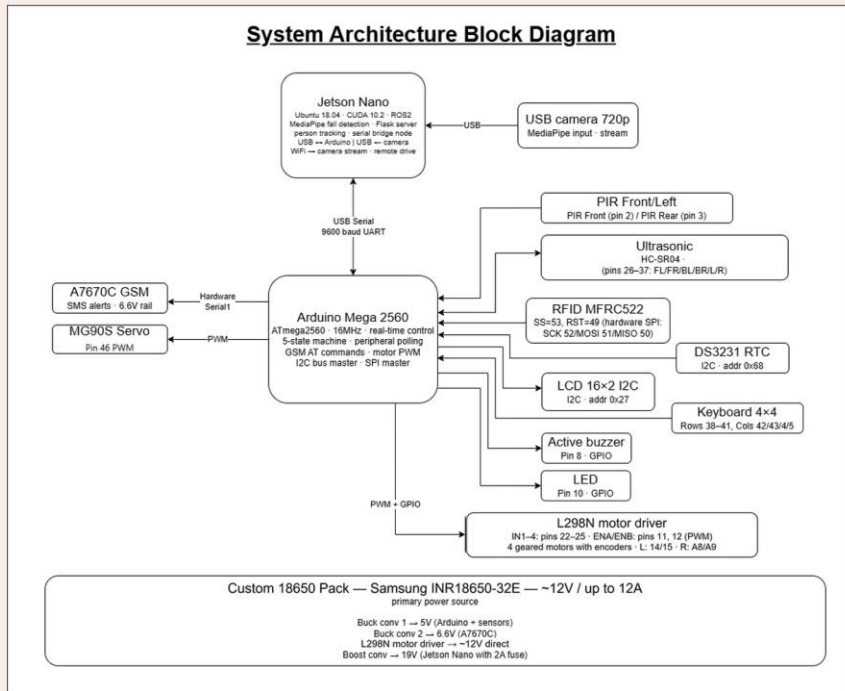
What trade-offs does a strict cost ceiling force?

5

Multimodal HRI at Low Cost

How can multimodal HRI be studied at this cost point?

Dual-Processor Architecture



Serial-linked dual-processor design separates safety-critical control from AI perception.

- **Arduino Mega 2560** — real-time motor control, obstacle avoidance, GSM alerts
- **NVIDIA Jetson Nano** — MediaPipe fall detection, person tracking, face recognition
- **GSM (LTE)** — all safety-critical alerts; no Wi-Fi dependency
- **Serial link** — inter-processor communication
- **Total BOM cost** — under \$500

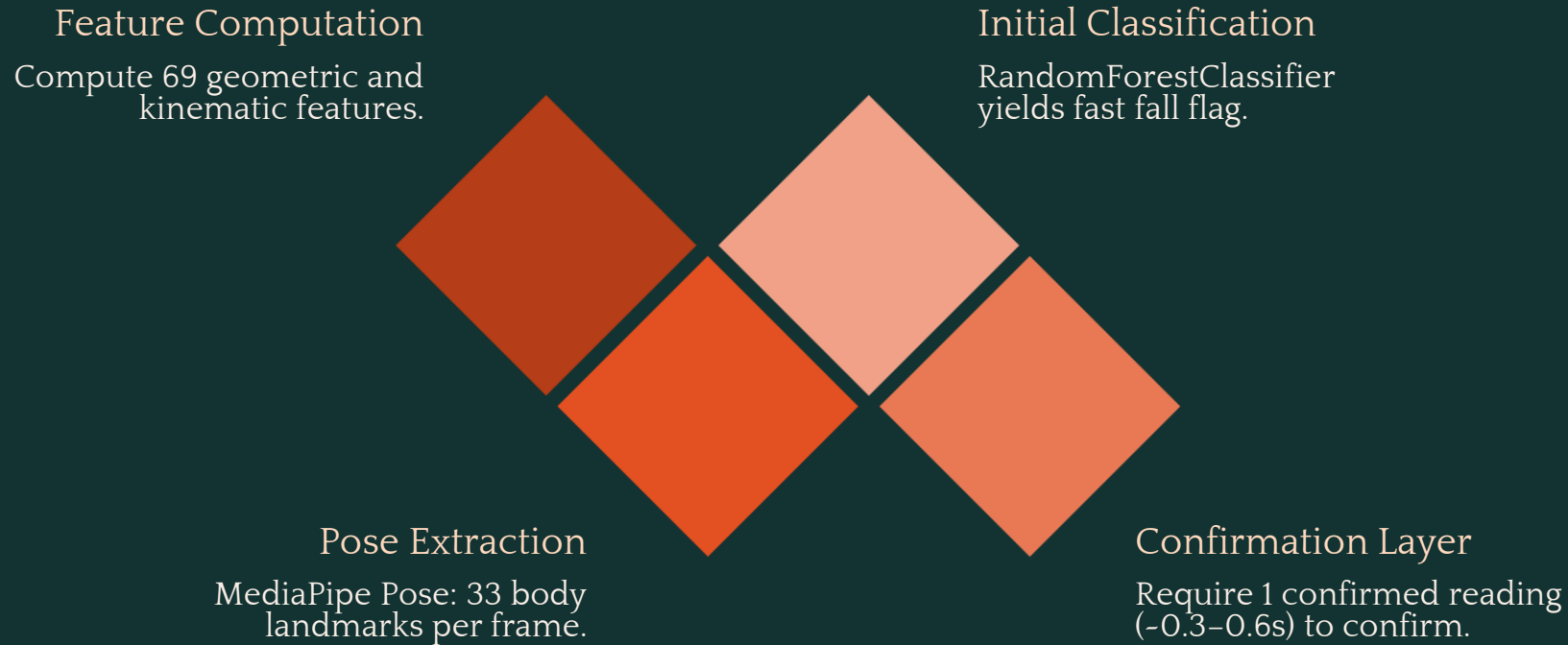


IMPLEMENTED FEATURES

15 Implemented Features

- Autonomous navigation/follow with obstacle avoidance
- AI-based fall detection (MediaPipe + RandomForest)
- GSM SMS alerts with delivery confirmation
- Medication reminders via RFID tag detection
- Live camera streaming
- Remote drive via web dashboard
- Companion greetings and face-recognition enrolment
- Color-band tracking fallback mode
- Person tracking
- Encoder-based closed-loop speed control
- Dual-motor feedback drive system
- Inbound SMS logging
- Two-layer fall confirmation logic
- Software I2C recovery fallback
- Serial inter-processor communication protocol

MediaPipe → Features → RandomForestClassifier



The two-layer confirmation — a fast initial flag, then one confirmed reading (-0.3–0.6 s) — favours quick response over longer verification, a deliberate speed-over-caution tradeoff for this use case.

Measured System Performance (Model Evaluation)

Fall-Detection Confusion Matrix

		Predicted		
		Fall	Not Fall	
Actual	Fall	55	6	87.6% Test accuracy
	Not Fall	10	58	89.6% Cross-validated
				90.2% Recall (fall class)

Fall Detection

- 87.6% test accuracy (target: 80%)
- 89.6% cross-validated accuracy
- 90.2% recall on fall class – prioritised over precision; a missed fall is worse than a false alarm

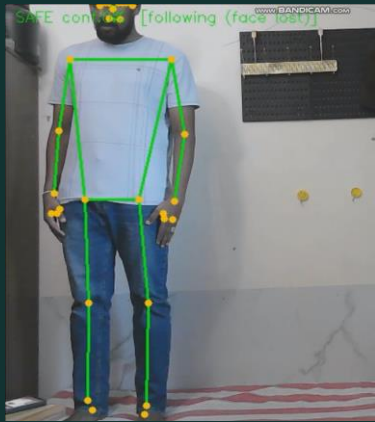
GSM Connectivity

- Full send, delivery confirmation, and inbound logging working end-to-end on replacement LTE hardware
- Original 2G module found network-incompatible – replaced during development

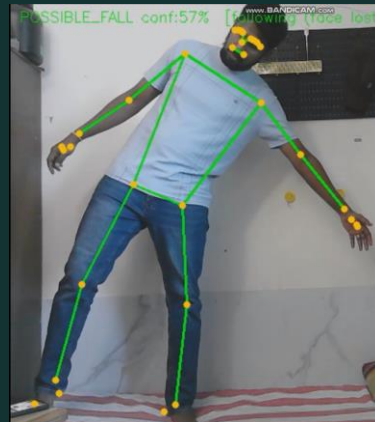
Motion Control

- 4-motor drive with dual encoder feedback; forward/reverse verified over repeated trials
- Obstacle-avoidance guard correctly halts the robot near obstacles

Fall Detection Tests



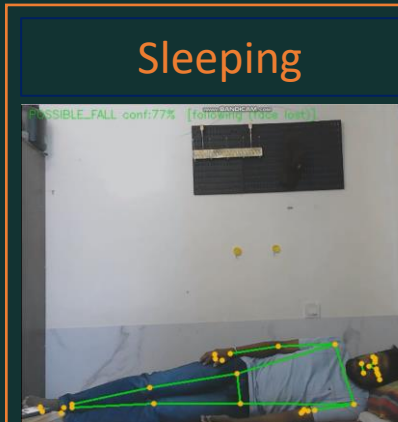
-Standing Status-
Possibility 0% < 80%
Alarm Off



-Fallen started-
Possibility 57% < 80%
Alarm Off



-Middle Status of falling-
Possibility 99% > 80%
Alarm On



-Sleeping Status-
Possibility 77% < 80%
Alarm Off



Fallen to ground (facing backward)

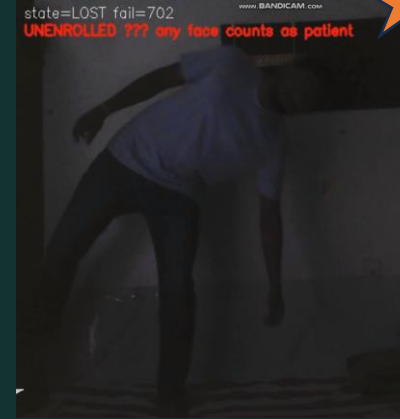
-Fallen Status-
Possibility NA ? 80%
Alarm Off



-Middle Status of falling-
Possibility 100% > 80%
Alarm On



-Bend down Status-
Possibility 97% > 80%
Alarm On



-Fallen Status (Very Low Light)-
Possibility > 80%
Alarm Off



Fallen to ground

-Fallen Status-
Possibility 89% > 80%
Alarm On



Fallen to ground (with mask)

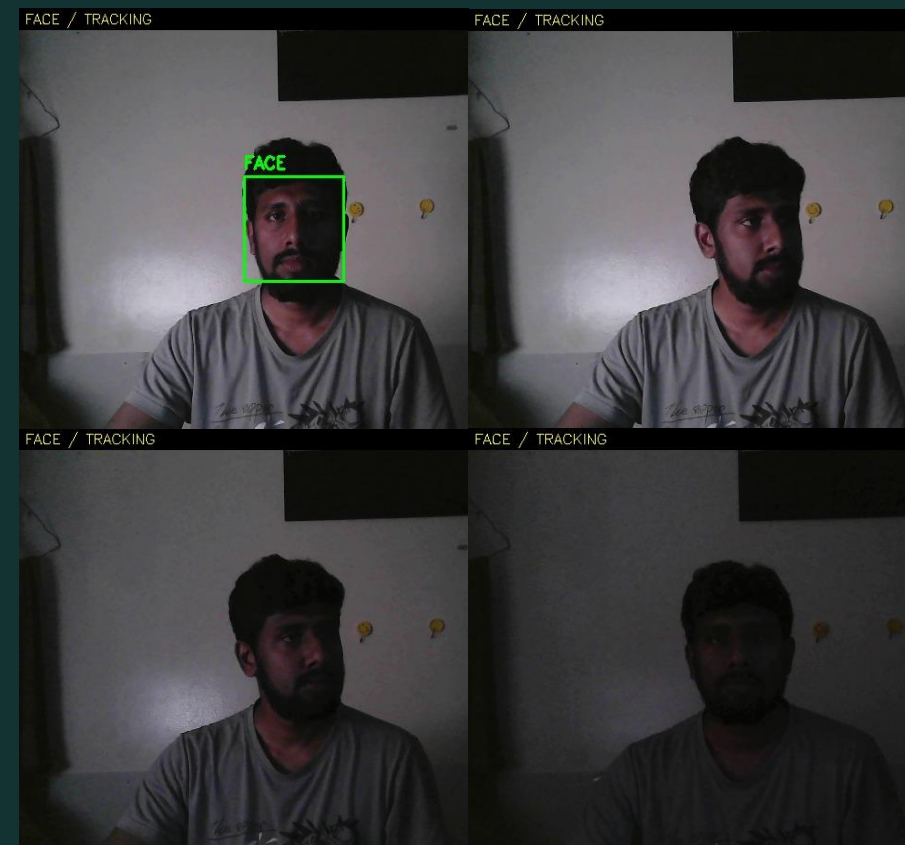
-Fallen Status-
Possibility NA ? 80%
Alarm Off

Person Detection Tests

Mid Light / Average Room Condition



Low Light / Minimum Room Condition / unbalanced light



KEY RESULTS

Band Detection Tests

Distance 50cm

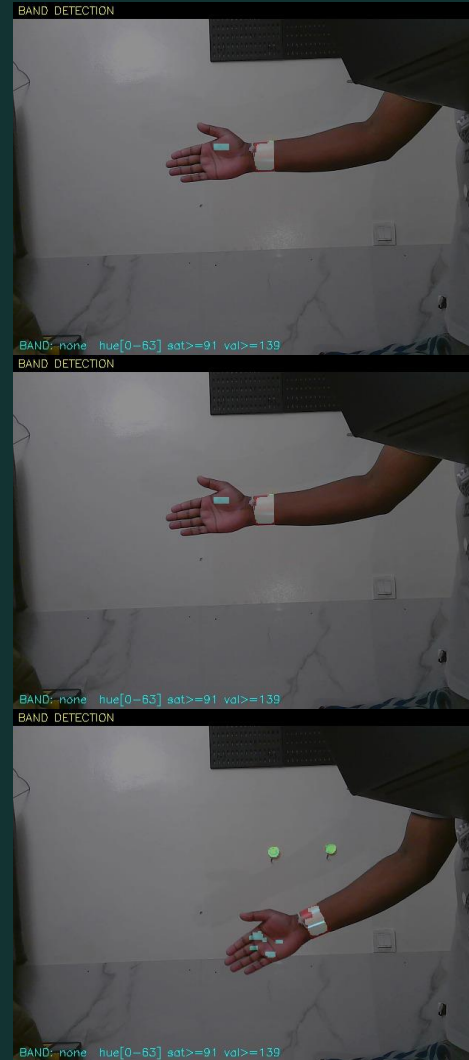


Detected

Not Detected

Detected

Distance 100cm

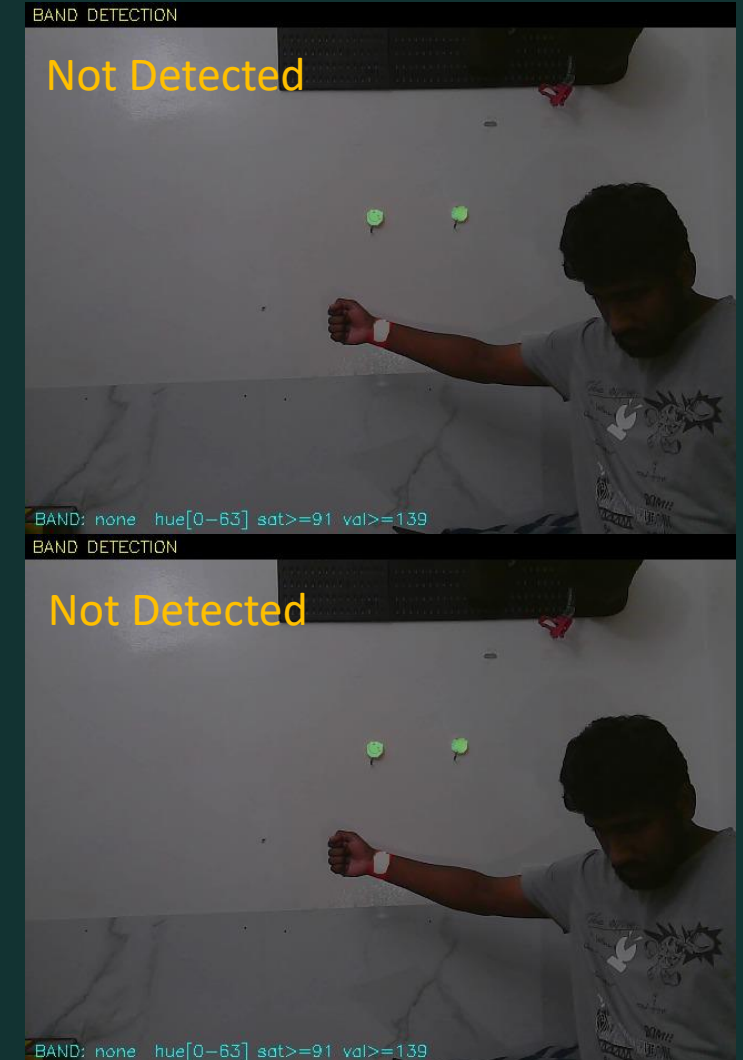


Not Detected

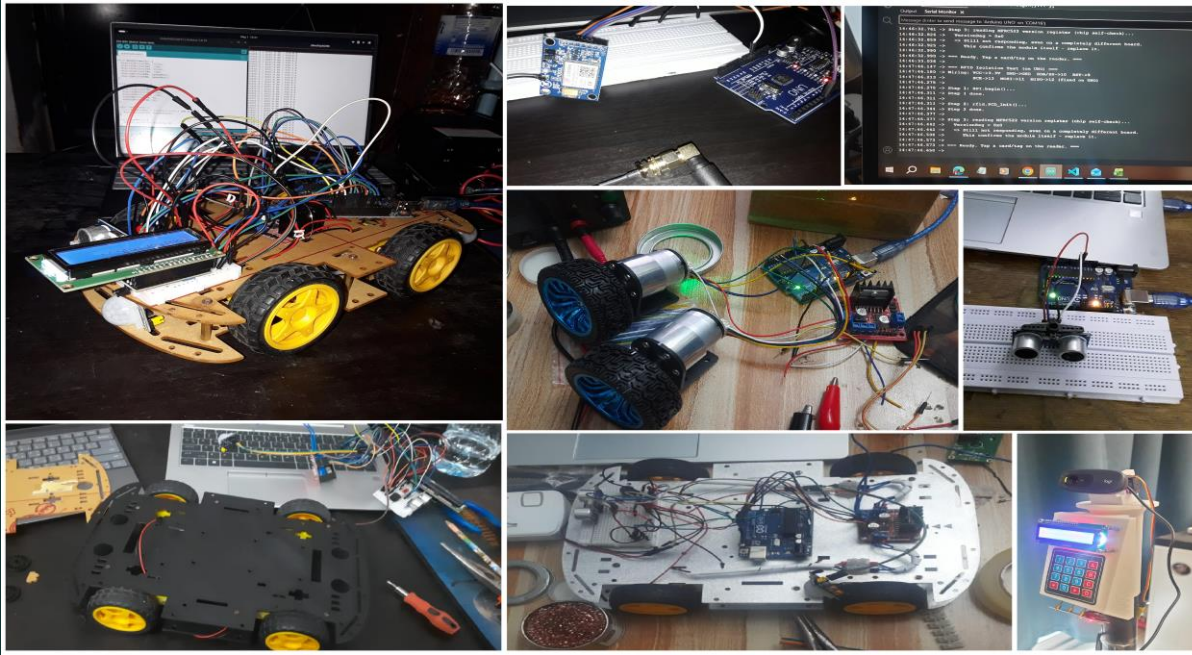
Not Detected

Not Detected

Distance 150cm



Systematic Fault Diagnosis as a Transferable Contribution



Two hardware investigations used the same repeatable elimination methodology – one a confirmed fault, one a verified power requirement – presented here as a transferable contribution, not merely a project-specific fix.

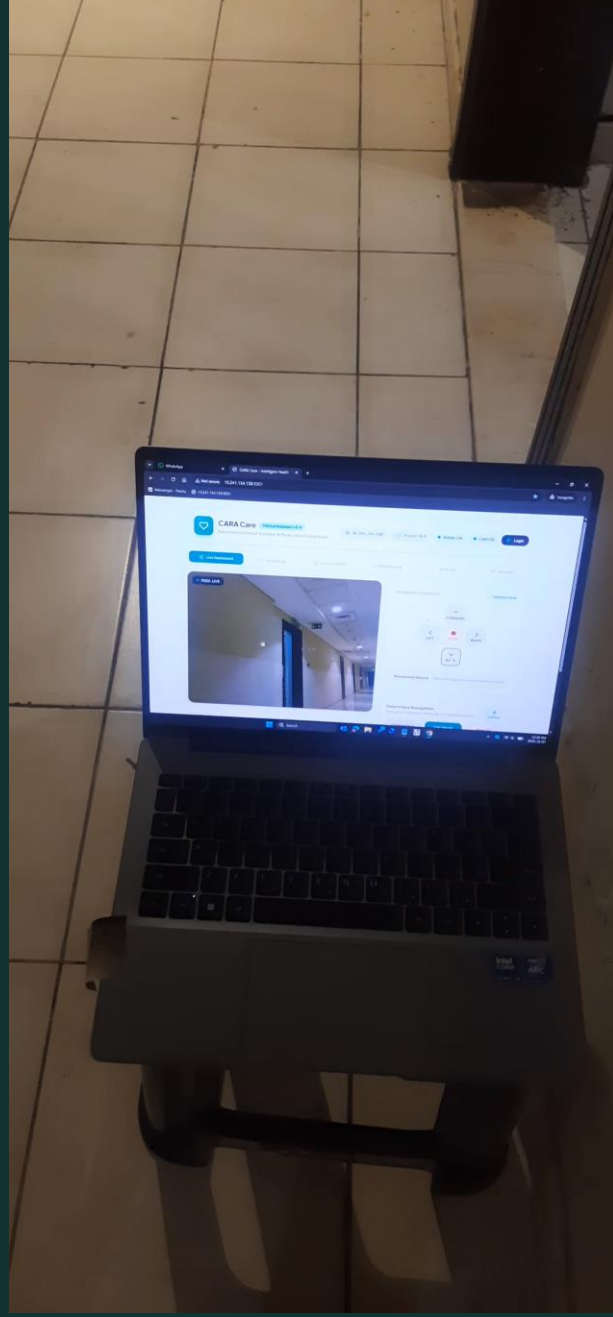
Finding – RFID Needs Clean Power

Isolated via systematic testing (battery, DC-DC voltage, capacitors, host device) that RFID needs clean, direct power – reads perfectly on laptop power, corrupted via the Jetson/battery rail. A deployment requirement, not a reader defect.

Fault 2 – Damaged I2C Pin

Physically damaged pin on the Mega recovered by migrating to a software I2C bus on alternate GPIO pins.

Full system testing outdoor



Medicine LogError LogSettings

FEED: LIVE



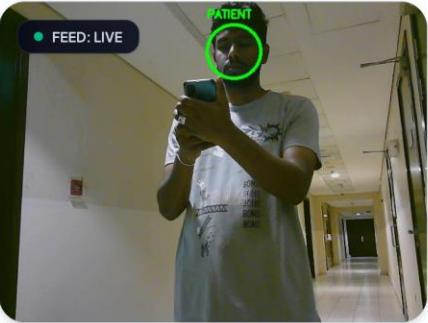
Medicine LogError LogSettings

FEED: LIVE



Medicine LogError LogSettings

FEED: LIVE



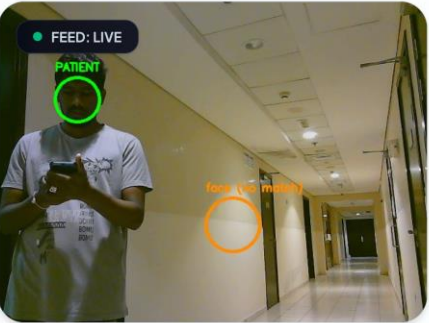
Medicine LogError LogSettings

FEED: LIVE



Medicine LogError LogSettings

FEED: LIVE



☒ Show face detection (green = recognized patient)
☐ Show skeleton

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☐ Show skeleton

TELEMTRY & DIAGNOSTIC STREAM

12:46:49 [drive] STOP ok
12:46:50 [drive] TURN_R ok

12:47

10.241.154.158Signal N/A

SERIAL ONCAM ONLogin

Live Dashboard

Action Log

Connected IPs

Medicine LogError LogSettings

FEED: LIVE



☒ Show face detection (green = recognized patient)
☐ Show skeleton

12:48

10.241.154.158Signal N/A

SERIAL ONCAM ONLogin

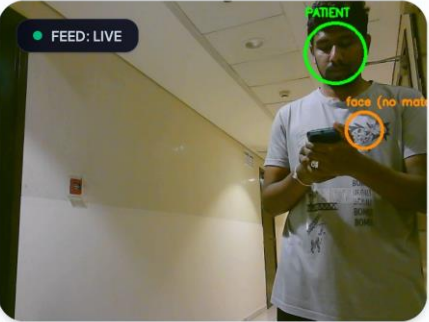
Live Dashboard

Action Log

Connected IPs

Medicine LogError LogSettings

FEED: LIVE



☒ Show face detection (green = recognized patient)
☐ Show skeleton

12:49

10.241.154.158:5001

Medication Companion

10.241.154.158Signal N/A

SERIAL ONCAM ONLogin

Live Dashboard

Action Log

Connected IPs

Medicine LogError LogSettings

FEED: LIVE



☒ Show face detection (green = recognized patient)
☐ Show skeleton

12:50

10.241.154.158:5001

Medication Companion

10.241.154.158Signal N/A

SERIAL ONCAM ONLogin

Live Dashboard

Action Log

Connected IPs

Medicine LogError LogSettings

FEED: LIVE



☒ Show face detection (green = recognized patient)
☐ Show skeleton

Active Link

LIMITATIONS

Honest Limitations

Single-Subject, Lab-Only Testing

All trials conducted with one participant in a controlled environment — real-world generalisation remains unvalidated.

In-Place Turning Fault

One drive side stalls under load during in-place turns; root cause identified as friction of the floor and unbalance of the robot.

Camera Tilt Servo

Confirmed dead in isolation on two separate pins — signal, power and ground all verified correct; a failed component, not a wiring or software fault. Identified as expected power for the motor is not received while requesting the service.

Multi-Person Face Recognition

Discrimination between multiple enrolled faces not yet formally tested.

Path to Commercial-Grade Reliability

01

Resolve Turning Fault

Diagnose and fix the asymmetric stall under load.

02

Formal Collision-Free Trial Count

Complete structured navigation trials with quantified success rates.

03

Clean Power Supply for RFID

Add a dedicated regulated rail so tag detection works reliably off battery/Jetson power, not just direct laptop power.

04

15-Feature End-to-End Trial

One continuous integrated demonstration of all features.

05

Ethical Clearance & Elderly Trials

Formal testing with elderly participants — the definitive validation step.

Thank You — Questions Welcome